

The Heart Manual

Post Myocardial Infarction
(Post MI) Edition



Put your heart attack
behind you with this
six-week programme
for healthy living

About the Heart Manual: Post MI Edition

Foreword

NHS Lothian's Heart Manual Programme has, over the years, become the leading home-based cardiac rehabilitation package in the UK and Ireland, contributing to the recovery and wellbeing of in excess of 600,000 patients.

To date, more than 5,000 healthcare professionals have also been trained as facilitators and the Programme has been refined to appreciate developments in acute clinical practice, including the more widespread use of angioplasty.

It is also used in a number of overseas countries, and reviewed and revised regularly in line with clinical guidelines, professional bodies and patient representatives.

The Programme is shown to be an effective method in promoting a range of clinical, psychological and service outcomes as specified in SIGN, NICE and international guidelines, and has been used as an exemplary model to adapt to other health conditions.

As always, the Heart Manual Management Group hopes you get the very best from the Programme and welcomes the views of users, their supporters, families and healthcare professionals. Please contact us with your comments.

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"The UK Heart Manual (NHS Lothian) is perhaps the most extensively studied self-management book for patients recovering from acute MI or coronary revascularization."

American Association of Cardiovascular and Pulmonary Rehabilitation,
the American Heart Association, and the American College of Cardiology, *Circulation*. 2019.

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The Heart Manual: Post MI Edition



This book is called the **Heart Manual: Post MI Edition** because it can help you to understand why you had a heart attack.

It will also tell you the things that you can do to recover in the quickest and safest way.

It can tell you about the medicines you are taking.

If you live with someone it can help them to understand your condition.

The main part of the book is divided into six weekly sections. There are some simple exercises to do, and each week there is a small section to read.

The bad news

The first reaction after a heart attack is often relief that you have survived. But once you get home, this may turn into worry about:

- how it will affect your life
- having another heart attack
- what you can or can't do
- going back to work.

Upset emotions are common. Most people in the same position occasionally feel:

- sad
- afraid
- bad-tempered
- moody.

The good news

These worries are often caused by:

- out-of-date ideas
- wrong ideas
- not knowing the true facts.

Reading this Manual will help you to find out the truth about heart attacks.

The upset feelings are normal and will pass as you get stronger.

Before you start the Heart Manual Programme please go to page 167 of this manual and complete your Baseline Questionnaire. You will be asked to complete a similar questionnaire at the end of the programme. This will allow you to compare how you are now, to how you are at the end of your Heart Manual Programme. This information is also very useful to us and we will ask you to return both questionnaires to the Heart Manual Department at the end of the programme.



How to use the Manual

Part 1: Your Heart Attack: the Facts

(pages 9–16)

This section is for you to read while you are recovering.

It is labelled like this:



It tells you the plain facts about your heart attack. You should read this section first, and you might want to come back to it a few times during your recovery.

If you live with someone, get them to read this section, too. It can help them just as much as it helps you.

You should listen to the 'Questions and Answers' CD or audio. You can access the audio online here: www.heartmanual.scot.nhs.uk/index.php/download/resources

Part 2: The Weekly Programme

(pages 17–132)

This part consists of six weekly sections.

The weeks are clearly labelled like this:



Each weekly section has important information to help you to recover, and an exercise programme to help you back to fitness in easy stages.

Every day:

- Follow your plan morning and evening as advised by your facilitator
- Each time mark your activities in the charts provided
- Practise relaxation by listening to tracks 1 and 2 on the relaxation CD or audio. You can access the audio online here:
www.heartmanual.scot.nhs.uk/index.php/download/resources

or scan this QR code



or download the HM Relax app via Apple or Google Play stores or scan the QR code. Please ask your facilitator for the username. The password is relax.



Part 3: Facts and Advice to Help Your Recovery

(pages 133–175)

This part provides extra information about your recovery, medicines and other things you might want to know. You can read it right through or dip into it to answer your questions.

The Heart Manual: Post MI Edition

Not just a book

This Manual isn't just a textbook of information. It is a proven rehabilitation tool – an important part of your treatment which has been demonstrated to help many people in the first weeks after a heart attack.

It is designed for you to use as you wish and for **you** to adapt to **your** needs, with regular support.

This Manual has been given to you by a healthcare worker (usually a **nurse**) who has been specially trained to work with patients using the Manual. We call this person a **facilitator**.

During the first 6 weeks at home, you will be contacted by another facilitator. He or she can discuss with you how you are getting on with the Manual, answer any questions you may have, and discuss future contact dates and times.

You might like to write down the names of your facilitators here:

Hospital facilitator

Name

Contact phone number

Community (home) facilitator

Name

Contact phone number



Important message

If you know someone else who has had a heart attack, **please do not show them this Manual or the CDs/audio.**

Do not advise them to do any of the exercises or give them advice from the Manual. You may be very tempted to help them, **but please don't. Instead of helping them you may harm them.**

This is very important because they may have a different or much more serious kind of heart condition. **Your advice may be wrong for them and you may cause them harm.**

There is no way that you can guess about this. This Manual is only suitable for people who have been advised to use it by trained healthcare staff.

None of the advice in this Manual is intended to replace any advice that you are given by your doctor.

Part 1

Your Heart Attack: the Facts

Have you completed your baseline questionnaire on page 167? If not please take a few minutes to complete it now. You will be asked to complete a similar questionnaire at the end of the programme. This will allow you to compare how you are now, to how you are at the end of your Heart Manual Programme.





Recovering

You and your heart attack

The first section of this Manual gives you the plain straightforward facts about your heart attack and answers some of the first questions that people usually ask. We suggest that you read through this section while you are still in hospital. If you live with someone, they should read it, too. If you don't receive the Manual until you are back home, read this first section now, and then move straight on to Part 2 (page 21) and start the weekly programme.

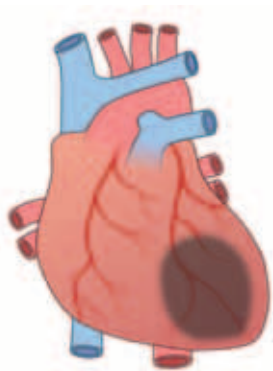
But please remember not to show it to anyone else who has had a heart attack.

This is because:

- **their heart attack may have been caused in a different way**
- **they may have a much more serious problem than yours**
- **the advice may be wrong for them.**

Only a healthcare professional can decide this.

What is a heart attack?



Heart attack

Many people believe that a heart attack occurs because the heart is 'worn out' or has 'packed up.'

This is completely untrue. The fact that you have had a heart attack doesn't mean that your heart is worn out or that you are 'finished.'

Your heart is a muscle which pumps blood around your body. Like all muscles, it has its own blood supply through arteries which run over its surface. These are called coronary arteries.

If one of these arteries becomes blocked suddenly, part of the heart muscle will be starved of blood and will become damaged. Doctors call this a **myocardial infarction**. Most people call it a **heart attack**.

Your initial treatment will depend on several factors, but mainly your symptoms, how long you have had them, and your test results. (e.g. ECG)

Restoration of blood flow is a key priority, and this can be achieved with intravenous drugs or by opening up the artery with a balloon catheter. (**ST elevation myocardial infarction**).

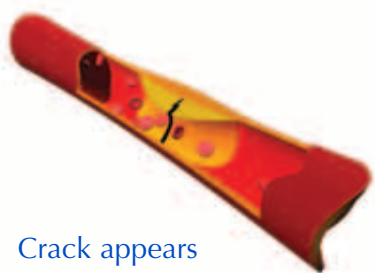
In other cases, blood thinning medication, including Aspirin, and/or other medications are introduced first, with further interventions as required. (**Non ST elevation myocardial infarction**).



What caused the blockage?



Narrowing



Crack appears



Clot develops

You may have been told that you have coronary artery disease. This means a gradual 'furring up', 'hardening' or thickening of the walls of the arteries, which makes parts of your arteries get narrower over a number of years. Many people have coronary artery disease without knowing it. Often it doesn't cause any harm.

Sometimes a narrow region of your artery may become damaged or inflamed. A small crack may appear in the lining of the artery where it narrows. When this happens, the body fights back by covering the damaged area with a small clot made of special blood cells called [platelets](#). It's rather like putting sticking plaster on a cut.

Usually the clot heals the damage, but sometimes the clot grows too large and blocks the whole width of the artery. If this happens, no blood can get past the clot, and as a result the heart muscle is starved of blood. This is a heart attack.

So in most cases a heart attack is caused by coronary artery disease – not by a problem with the heart.

You will find out much more about [coronary artery disease](#) – **and what you can do about it – in Part 2 of the Manual.**

Where is the blood clot now?

If you were given thrombolysis – a clot-busting drug, such as streptokinase or alteplase – the clot has probably dissolved and disappeared already. If not, don't worry. Your body's natural clot-busters will be getting rid of it over the next few weeks. In the mean time, it is unlikely to do you any harm.

The reason why doctors try to dissolve the clot with drugs is to prevent any more damage to your heart muscle. They can only do this within a few hours after the heart attack, and only if they are sure that the heart is being damaged.

This can be confirmed with specific blood tests and a scan called an echocardiogram.





Where is the blood clot now? (continued)

Angioplasty (stent)



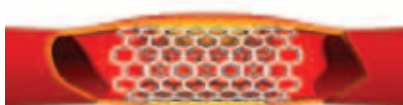
Catheter



Balloon inflated



Balloon deflated



Stent

Your cardiologist may have decided to treat your heart attack in another way, and may have asked your permission to perform a percutaneous transluminal coronary angioplasty (PTCA) also referred to as a percutaneous coronary intervention (PCI). Many people simply call it angioplasty or stenting.

A special type of catheter that has a small balloon on the end is threaded through an artery in your arm or at the top of your leg and passed up into the coronary arteries.

Using an X-ray camera, the balloon is positioned in the narrow part of the artery. It is then inflated with enough pressure to stretch the artery where the coronary disease and clot have narrowed it. The balloon is deflated and dye is injected to see whether the narrowing has been stretched wider. The balloon may need to be inflated several times in order to get the best result.

The cardiologist may decide to put a small metal tube called a **stent** inside the artery to make sure that it stays open after the angioplasty. The stent can be stretched by the balloon to fit the exact size and shape of the inside of the artery. Once it has been stretched it acts like scaffolding to hold the artery open while it repairs itself. Eventually the stent becomes embedded in the lining of the artery, helping to prevent it from becoming narrowed again in future.

In some cases there may be more than one area of narrowing in a coronary artery or other arteries may be affected. In the event of a heart attack the cardiologist will treat the narrowing thought to be the cause of the coronary event.

It may be necessary to have further procedures in the future if you have ongoing symptoms of angina.



Is that why I was in Coronary Care?

It's one reason. The other reason is that the first 24–48 hours is the most unstable time.

Your heart has a natural 'pacemaker.' It works by sending electrical impulses that cause the heart to beat (this gives rise to your pulse). While the heart attack is happening, your natural pacemaker may not function correctly. This can make the heart beat too slowly, too fast or go into spasm. In some cases the heart stops beating effectively or 'fibrillates.' This may lead to a cardiac arrest.

When this happens you may black out. The simplest treatment is to give the heart muscle an electric shock. Without that treatment you may die, but if you are treated quickly you are likely to make a full recovery.

That is why it is so important not to delay going to hospital if you are having chest pains.

After a day or so the risk of the heart beating abnormally goes down, and you can come off the heart monitor.

Will my heart recover?

The good news is that it's well on the way to recovery already. When the heart muscle is damaged, it becomes swollen and painful. However, after a few hours the body starts to make repairs and the pain settles down.

The healing continues for several weeks, until all that is left of the heart attack is a scar.

Imagine that you have just had an operation. Today you would have a red wound, but 3 months from now all you would see would be a small scar. It's the same process after a heart attack.

Things to remember



- The heart has miraculous powers of recovery.
- It is the strongest muscle in your body.
- No other muscle has the ability to work non-stop 24 hours a day.
- Because it works all the time it is extremely fit.
- It is also capable of doing much more work than most of us ever need.

Some general information

- **Driving:** The advice given about returning to driving varies depending on your diagnosis, treatment and the type of driving licence you hold. All drivers should let their insurance company know about their condition to ensure they are covered if they need to make a claim. Before returning to driving it is best to get some advice from your doctor or discuss it with your facilitator. Legal guidelines change from time to time so it is important you get the correct information. You can also check the latest information on driving and medical conditions at www.gov.uk.
- **Going back to work:** For advice on going back to work if you are employed, read page 129 in Part 2 of the Manual.
- **Sex:** For advice on sex, read pages 117–118 of the Manual.
- **Flying:** Recent guidelines suggest that the majority of patients with heart conditions can travel safely, although some may require additional assistance. It is important to check with your GP and facilitator before planning any trips and you are advised to inform your insurance company when organising anything. For further information on fitness to fly you can consult the Civil Aviation Authority website www.caa.co.uk.

Remember to make sure that you have the right amount of medication for the duration of your trip and always carry it in your hand baggage. Make sure you take advice if you are going to miss any hospital appointments.



Will I have another heart attack?

There are no guarantees for anyone, whether they have had a heart attack or not

Many people think that once you have had one heart attack, all you can look forward to from then on is one heart attack after another until you die.

This is completely wrong.

It is very unusual for a person to have a whole series of heart attacks.

The majority of people who take the warning and make some important changes to their lifestyle never have another heart attack.

Why ... ? How ... ? When ... ?

You probably have many more questions to ask. It may help to write them down here and ask them the next time you speak to your doctor or facilitator. Write down the answers, too. Don't be afraid to ask what you think is a 'silly question.' It is important to get questions cleared up so that you don't dwell on them. Your doctor or facilitator will have heard it before!

Questions

Answers



Part 2

The Weekly Programme

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Week 1

Getting Home - Getting Better



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